POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks : 100
Course: Database Management System (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is data abstraction in DBMS? Explain in detail. 7
   b) Consider you're asked to design a database for the Exam section of your college. Draw its ER diagram assuming required entities and their attributes. 8

2. a) Convert the ER diagram that you designed in question no 1 b) into relational schema. 7
   b) What are the views? Consider the table **tbl_emp** as follows: 8

Emp_id	Emp_name	Salary	Department	Date_of_joining
101	Anish	20000	Packing	2070-02-01
102	Rojina	18000	Cleaning	2075-04-06
103	Sita	35000	Polishing	2078-09-12

   Write the SQL statements for the following:
   i. Create the above table by considering Emp_id as primary key and insert above records.
   ii. Change the department of Anish to marketing.
   iii. Increase the salary of employees whose department is Cleaning by 12%
   iv. Find the name of employees having salary greater than 16000 and who joined after 2072-11-25/
   v. Add a new column Address to the above table.
   vi. Delete the entire table.

   OR

   Why is joining in SQL necessary? Explain inner join, Natural join, and Outer Join with suitable examples.

3. a) What are the different types of integrity constraints? Explain with examples. 8
   b) What is denormalization? Why is it necessary? Explain in detail. 7

4. a) What is the role of views in security? Explain the concept of authorization with suitable examples. 7
   b) Explain the basic steps in query processing in detail. 8

5. a) Explain how the records are organized using the fixed-length and variable length records. 7
   b) What is serializability and why is it needed? Explain the ACID properties in brief. 8

github.com/nanish4dev

6. a) What is crash recovery? Explain log-based recovery method with example. 7

OR

What is transaction rollback? Explain how the Remote Backup System provides high availability and recovery facility.

b) List out the different categories of NoSQL databases. Explain the concept of blockchain with its properties. 8

7. Write short notes on: (Any two) 2x5
   a. Nested Queries
   b. Third Normal Form (3NF)
   c. Lock-based protocols for concurrency control

github.com/harsh4dev

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year: 2023
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Attempt all the questions.

1. A) Differentiate between database schema and instances. Briefly describe DDL, DML, and DCL. [8 marks]

1. B) Define relational data model. Draw an E-R diagram for a Library Management System Including primary key, weak entity, composite attribute, derived attribute and multivalues attributes in your ER diagram. [7 Marks]

2. A) Suppose we have the following relation. [8 Marks]

Employee(person_name, street, city)
Works(person_name, company_name, salary)
Company(company_name, city)
Write relational algebraic expressions for the following queries:

i. List the name and city of employee who work in "pokhara" and have salary greater than rs.50000.
ii. Find the names of all employees who work for "ABC bank".
iii. Delete all the employee who come from "Chitwan".
iv. Increase salary of all employee by 15%.

2. B) What are different kinds of joins? Explain in brief. [7 Marks]

3. A) Write SQL statements for the following queries using the given Employees relation: [8 Marks]

E_id	Fname	Lname	Department	Salary	Hire_Date
01	Ramu	Bashyal	Sales	20000	2023-08-08
02	Damu	Pandey	IT	50000	2022-01-01
03	Biru	B.K.	Sales	40000	2021-02-10
04	Hiru	Dhamala	HR	35000	2023-12-18
05	Biren	Khadka	IT	60000	2012-10-22

i. Create a database named Company and Employees relation.
ii. Create a view that shows E_id, Department, and Hire_Date of all employees
iii. Modify the table such that the Department of Biren is HR now
iv. Delete the record of employees whose Lname is "Pandey".

3. B) What is referential integrity? Explain about trigger with an example. [7 Marks]

4. A) What is database normalization? Explain in detail about 1NF, 2NF, 3NF with suitable examples. [8 Marks]

4. B) What are authorization and authentication? Why are they important? Exlpain in detail. [7 Marks]

5. A) What are the steps in query processing? Make an operator tree for the following SQL expression:

```
Select customer_name
FROM branch, account, depositor
WHERE branch_city = 'btl' AND balance>2000;
```

5. B)What are the benefits of using B Tree index over the sequential and indexed sequential file organization? Explain. [8 Marks]

6. A) Explain log based recovery system with an appropriate log record example. [7 Marks]

6. B) Why should the transactions schedule be serialized? Explain conflict and view serializability with an example. [8 Marks]

7. Write short notes on: (Any two) [2x5 Marks]

a) Data Dictionary
b) Stored procedure
c) Object oriented database

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1. a) Define schemas and instances. What are the major functions of Database Manager? Discuss. [7 Marks]

1. b) Explain the benefits of data model. Construct an ER diagram for keeping records for Library Management system. [8 Marks]

2. a) Consider the following schema: [8 Marks]

EMPLOYEE(Employee_ID, Employee_Name, Department)
PROJECT(Project_ID, Project_Title, Budget)
WORKS_ON(Employee_ID, Project_ID, Hours_Worked)
Write the relational algebra expression to:

i. Find the names of all employees who are working on a project with a project title that contains the word 'Development'.
ii. Update the budget of projects to 1,000,000 where the budget is currently less than 500,000.
iii. Find the total hours worked by each employees across all projects.
iv. Update the name of 'HR' department to 'Human_Resources'.

2. b) Write the SQL statements for the following queries by reference of Hotel_details relation. [7 Marks]

3. a) What are the advantages of normalization? Illustrate importance of normalization with suitable examples. [8 marks]

3. b) Why Integrity constrains are used? Explain types of integrity constraints with examples. [7 Marks]

4. a) Differentiate between authorization and authentication. Explain about private key cryptography with example. [8 Marks]

4. b) What is query processing? Explain its steps with block diagram. [7 marks]

5. a) Create a B+ tree of order 4 with the following data: (1,4,7,10,17,21,31,25,19,20,28) [7 MARKS]

5. b) Explain log based recovery and different methods of keeping log along with examples. What is shadow paging? [8 marks]

6. a) What is view serialize ability? Explain 2PL and time stamping. [7 marks]

6. b) What are distributed database systems? Explain advantages and disadvantages of object oriented database model. [8 marks]

7. Write short notes on: (Any two) [2x5 Marks]

a) Data Abstraction
b) ACID
c) Triggers

github.com/harsh4dev

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Attempt all the questions.

1. a) Differentiate between schema and instances. Explain data independence and its importance. [8]

b) Design an ER diagram for an Online Food Delivery System to manage customers, restaurants, orders, and delivery. Draw the ER diagram with cardinalities and constraints. [7]

2. a) What is relational algebra? Explain Selection, Projection, Union, Cartesian-product, Join and Set-Difference operations with suitable examples. [7]

b) Consider the given relational schema:

Movie(Title, Director, Myear, Rating)
Actors(Actor, Age)
Acts(Actor, Title)
Directors(Director, DirAge)

Write the SQL statement for the following task:

i. Find movies made by Gopal Varma after 1997.
ii. Find all actors and directors.
iii. Find number of movies which are directed by Devid Dhaban having actor. "Bishnu"
iv. Find (Director, Actor) pairs where the directors is younger than the actor.
v. Find the number of movie actor wise.

3. a) Explain functional dependency with its types. For R = {A, B, C, D} and F = {A→BC, B→C, AB→C, AC→D}, List the closure of functional dependency F. [7]

OR

What are database constraints? Explain its types with suitable examples. When do we need Denormalizations? Explain.

b) Define normalization. Create relation in Unnormalized Form (UNF) and normalize it up to the Second Normal Form (2NF). [8]

4. a) Why security is needed in database? Explain access control mechanisms. [7]

OR

As a database administrator, purpose the measures to protect the database against physical and cyber threats.

b) Define ACID properties. Explain view serializability with suitable examples. [8]

5. a) What do you mean by query processing? Explain the steps of query processing with necessary diagram. [7]

b) Construct a B+ tree for the following set of key values: {2, 5, 7, 9, 11, 17, 21, 23, 29, 31}. Assume that the tree is initially empty and values are added in ascending order where the pointer number is FOUR. [8]

6. a) Consider following two transactions T1 and T2 given in figure where T1 executes before T2. Also consider initial values of A, B, and C are 1000, 2000, and 3000 respectively. [7]

T1	T2
Read (A)	Read (C)
A = A – 500	C = C – 200
Write (A)	Write (C)
Read (B)	
B = B + 500	
Write (B)	

Use the concept of deferred and immediate database modification log-based recovery method in given problem.

7. Write short notes on: (Any Two) [2x5]

a. Query Cost Estimation
b. Set Operations in SQL
c. NoSQL Databases and its characteristics.

Github.com/theharshchaudhary

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		Time	: 3 hrs.

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Attempt all the questions.

1. a) Describe the architecture of a Database Management System and Database application (1-tier vs 2-tier vs 3-tier). [7]
b) Create an ER diagram for a movie rental store where: [8]
i. Customers rent movies
ii. Movies have movie ID, title, genre, and release year.
iii. Each customer has a customer ID, name and membership type.
iv. Capture the rental date and return date for each movie rented.

2. a) Consider the following relational database where the primary keys are underlined: [8]
Empdetail (emp-name, street, city)
Jobdetail (emp-name, university-name, salary)
Universitydetail (university-name, city)
Headdetail (emp-name, head-name)
Express the following queries in relational algebra:
i. Give all employees of Pokhara University a 15% salary raise.
ii. Update the city of Amit to Butwal
iii. Give all managers in this database a 25 percent salary raise
iv. Find all employees name who work for Pokhara University and has salary greater than 70000
b) Consider the relational database with the following relations: [7]
Student (StudentID, Name, DeptID, Age)
Department (DeptID, DeptName, HOD)
Enrollment (EnrollID, StudentID, CourseName, Grade)
Now write the SQL Command for the following:
i. Write SQL statements to create the student table.
ii. Display each student's name, department name, and course name for all enrolled students.
iii. Display the average grade for each course, based on the Enrollment table and group the result by course name.
iv. Update the age of student 'Iris' to 26.

3. a) What is Domain, Entity Integrity, and Referential Integrity constraints? Explain with examples. [8]

OR

Define normalization and explain its purpose in database design. Discuss the problems (anomalies) that occur in unnormalized database.
b) What is relational model? What are the features of good relational database design? Explain. [7]

4. a) Explain access control authorization, and views in database security. [7]
b) What is an execution plan? Describe how equivalence rules are used in optimizing queries in DBMS. [8]

5. a) Define indexing. Differentiate between B+ Tree Indexing and Hash Indexing. [7]

OR

Discuss the organization of records in Heap and Indexed Sequential file systems.
b) What is Serializability? Explain the lock-based concurrency control. [8]

6. a) What is transaction log? How are undo and redo operations handles using log records? Explain. [8]
b) Discuss the advantages and limitations of NoSQL databases. Explain different categories of NoSQL. [7]

7. Write short notes on: (Any Two) [2x5]
a. Data Independence
b. DDL and DML
c. Denormalization

Github.com/theharshchaudhary

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Attempt all the questions.

1. a) What is a database management system? Difference between Physical and Logical Data Independence. **[7]**
b) Draw E-R diagram for airline reservation system. The system must keep track of customers and their reservations, flights and their status, seat assignments on individual flights. Include appropriate relationship and cardinality. **[8]**
OR
Draw an ER diagram of an "Exam System of Pokhara University" using extended features. Explain Primary key and foreign key.

2. a) Consider the following schema: **[7]**
STUDENT(Student_ID, Student_name, Major)
COURSE(Course_ID, Course_name, Credits)
ENROLLS_IN(Student_ID, Course_ID, Grade)
Write the relational algebra expressions for the following cases:
i. Find the names of all students who are enrolled in courses with names that start with "C".
ii. Update the grades of students to 'A' whose current grade is 'B'.
iii. Find the average credits of courses each student is enrolled in.
iv. Update the name of the course "Physics" to "Advanced Physics".
b) Let us consider the following relation: **[8]**
Sailors (<u>sid</u>, sname, rating, age)
Boats (<u>bid</u>, bname, color)
Reserves (<u>sid</u>, <u>bid</u>, day)
Write a SQL statements for the following:
i. Find the records of sailors who have reserved boat number 75 (bid=75).
ii. Update the color of the yellow boat to green.
iii. Delete the records of sailors whose rating is less than 5.
iv. Find the name and rating of sailors who have reserved a black or blue boat.
v. Find the age of sailors who have reserved boat number 25 on day 3.

3. a) Define integrity constraints? Explain different integrity constraints with suitable examples. **[8]**
OR
Why is database normalized? Explain about 1NF, 2NF, 3NF & BCNF.
b) Explain different types of functional dependencies with examples. Find the closure of attributes A, AC and ABC for the relation R (A, B, C, D) where A→B, BC→D are the dependencies in R. **[7]**

4. a) Compare authentication and authorization. Explain the different types of access control mechanisms. **[7]**
b) How query optimization is carried out? Explain about cost estimation of query. **[8]**

5. a) Construct a B+ tree of order 4 for the given key values assuming empty tree initially and keys arranged in ascending order: **[7]**
{4, 8, 1, 20, 5, 15, 16, 7, 9, 38, 34, 6, 39, 25}
b) What is serializability? Explain how concurrency control protocols are used? **[8]**

6. a) What are NoSQL databases? Explain the advantages of NoSQL databases. **[7]**
b) Differentiate between centralized and distributed databases. Explain in brief about NOSQL. **[8]**

7. Write short notes on: (Any two) [2 × 5]
a) Equivalence of expressions
b) Stored Procedures
c) ACID Property

Github.com/theharshchaudhary